

Chemistry Ch 4.3: Periodicity of Properties

Chapter 4 · kitab ke page 45 se 72 · sections 4.3.1 se 4.3.5, paanchon rujhanat

Paanch istilahein

Shielding effect: andar wale electrons bairooni electron ko nucleus ki kheench se **bachate** hain.

Atomic size: atom ka nasf qutar (radius).

Ionization energy: bairooni electron **nikalne** ke liye darkar energy.

Electron affinity: electron **lene** par kharij hone wali energy.

Electronegativity: bond ke electrons ko apni taraf **kheenchne** ki salahiyat.

Paanchon ka rujhan

Property	Period mein (bayein se dayein)	Group mein (ooper se neeche)
Shielding effect	Taqreeban wohi	Barhta hai
Atomic size	Ghatta hai	Barhta hai
Ionization energy	Barhti hai	Ghatti hai
Electron affinity	Barhti hai	Ghatti hai
Electronegativity	Barhti hai	Ghatti hai

Paanch yaad karne ki zaroorat nahin, sirf ek

Sab kuch **atomic size** se nikalta hai, ulta.

Atom **chhota** = bairooni electron nucleus ke **qareeb** = kheench **zyada** = nikalna **mushkil** aur lena **asaan**.

Yani ionization energy, electron affinity aur electronegativity **teenon barh** jati hain.

Period mein size ghatta hai, to teenon barhti hain. Group mein size barhta hai, to teenon ghatti hain. **Bas size yaad rakhein, baqi khud nikal aayega.**

Quiz

1. Define **shielding effect**, **ionization energy** and **electronegativity**.
2. Copy the trend table for all five properties, across a period and down a group, and explain in one or two sentences why ionization energy rises across a period.